

ExampleHR – Time-Off Frontend

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NEXT.JS 16

REACT 19

TS

TYPESCRIPT 5

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STORYBOOK 10

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PLAYWRIGHT

Take-home assignment: a time-off request management frontend that presents balances and manages request lifecycles while staying honest about the fact that the HCM system (the source of truth) owns the data.

Stack

Layer	Technology
Framework	Next.js 16 (App Router)
UI	React 19
Language	TypeScript 5
Component Dev	Storybook 10
Testing	Vitest + Playwright
Mock API	Next.js Route Handlers
State	React Context + useReducer

Getting Started

```
npm install
npm run dev           # http://localhost:3000
npm run storybook     # http://localhost:6006
npm test              # unit tests
```

Vercel deploy: examplehr-timeoff-tau.vercel.app · Storybook: [/storybook/](http://localhost:6006)

Project Structure

```
src/
├─ app/
│   ├─ api/hcm/                # Mock HCM endpoints
│   │   ├─ balance/           # GET single balance
│   │   ├─ balances/          # GET batch balances
│   │   └─ requests/          # CRUD time-off requests
│   ├─ layout.tsx
│   └─ page.tsx                # Main page with role switching
├─ components/
│   ├─ BalancesTable.tsx      # Employee balance view
│   ├─ RequestForm.tsx        # Time-off submission form
│   ├─ RequestList.tsx        # Request list
│   ├─ ApprovalCard.tsx       # Manager approve/deny card
│   ├─ EmployeeView.tsx       # Employee view orchestrator
│   ├─ ManagerView.tsx        # Manager view orchestrator
│   └─ Notification.tsx       # Toast notifications
└─ lib/
    ├─ types.ts               # Shared types
    ├─ mock-hcm-data.ts       # Mock HCM data store
    ├─ hcm-client.ts           # API client
    └─ store.tsx              # State management
```

Key Design Decisions

- **Optimistic updates with rollback** – UI responds instantly; if HCM rejects, the optimistic state is rolled back with a clear error notification.
- **Periodic reconciliation** – balances are polled every 30s to catch external mutations (work anniversaries, new year rollovers). A stale indicator shows freshness.
- **Defensive HCM handling** – mock HCM simulates random failures (10%), silent failures, insufficient balance rejections (5% conflict), and variable latency.
- **Role switching** – toggle between Employee and Manager views to see both sides of the request lifecycle.

UI States Covered

Stories for: loading, empty, error, stale, live, refreshing, submitting, insufficient balance, no balances, optimistic, and conflict states.

Technical Requirement Document

See [docs/TRD.md](#) for the full architecture decision record, alternative analysis, and test strategy.

Deliverables

- ☒ TRD
- ☒ Source code
- ☒ Storybook — local: `npm run storybook` · deployed: [/storybook/](#)
- ☒ Unit tests (12 passing)
- ☒ Browser interaction tests

Screenshots

ExampleHR
Time-Off Management

EmployeeManager View

Your Balances

Synced at 16:23:00

LOCATION	TOTAL DAYS	USED	AVAILABLE
New York	20	6	14
London	20	6	14
Tokyo	20	3	17

Request Time Off

Location

New York (14 days available)

Available: 14 days

Days Requested

1

Submit Request

Your Requests

Alice Johnson

New York - 2 days

Pending

05/06/2026

ExampleHR
Time-Off Management

ManagerEmployee View

Pending Approvals (2 requests)

Alice Johnson

Location

Days Requested

Available Balance

Submitted

Deny

Approve

New York

2

18 days

03/06/2026

Bob Smith

Location

Days Requested

Available Balance

Submitted

Deny

Approve

London

3

15 days

02/06/2026

Storybook

Find components

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COMPONENTS

ApprovalCard

Pending With Balance

Pending Unknown Balance

BalancesTable

Notification

RequestForm

RequestList

VIEWS

ManagerView

Alice Johnson

Location

Days Requested

Available Balance

Submitted

Deny

Pending

New York

3

15 days

03/06/2026

Controls 2

Actions

Interactions

Visual tests

Accessibility 1

Name	Control
request	request: { id: "req-1" employeeId: "emp-1" employeeName: "Alice Johnson" locationId: "loc-1" locationName: "New York" daysRequested: 3 status: "pending" createdAt: 2026-06-03T19:29:50.321Z